

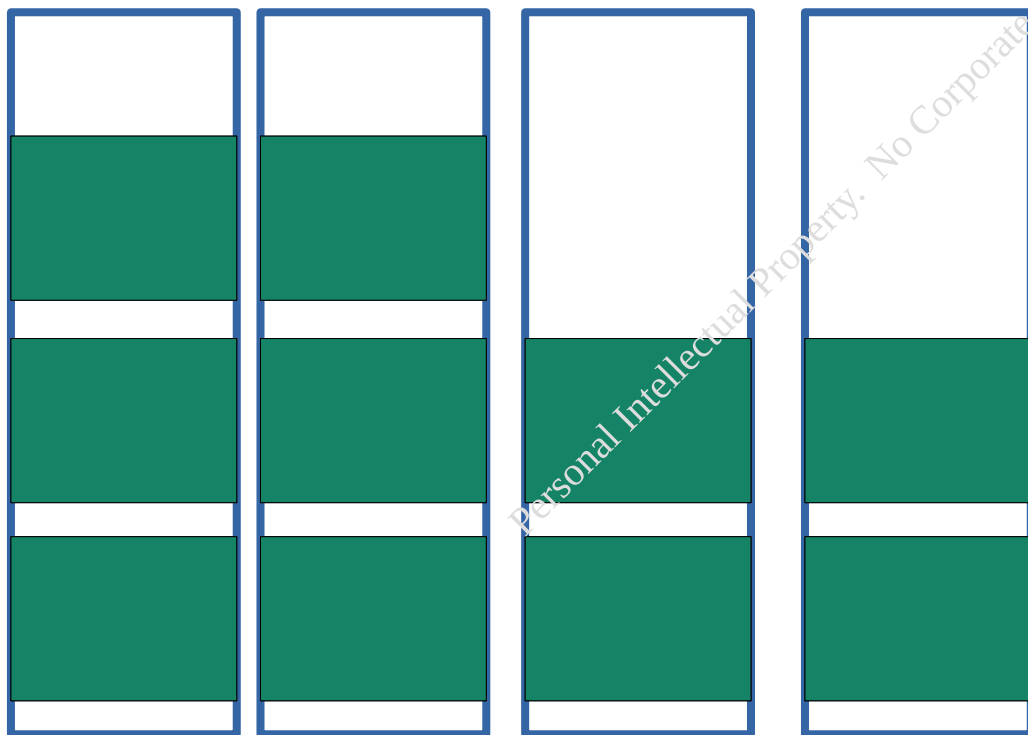
Existing setup/configuration as inventoried/investigated

Ranges for typical/peak loads are best guesstimate,
actual loads needs validation.

Nearly all physical setup has observed average between 2-12kw, but platform is capable of ~15-20kw peak depending on gpu workload.

A few systems with the 6-8 populated psu, but only 4-5 connected for power, maybe 1-2 have 8 psu connected.

Current psu population and cabling (less than 6-8 psu) may not satisfy intended redundancy models.



Per system/platform documentation and best practices:

The server supports configurations with six or eight power supplies:

- Six power supplies: Supports N+1 redundancy with OVS.
- Eight power supplies: Supports N+1 or N+N redundancy with OVS.

Following is the list of supported type:

- 2600-watt (or 3200w) Titanium, input power 200-240 Vac

Important: Power supplies and redundant power supplies in the server must be with the same power rating, wattage or level.

Minimal configuration for debugging

- System board assembly that includes the processor board and system I/O board
- Firmware and RoT security module
- Two processors
- Twenty-four memory modules
- Six or eight power supplies, depending on the configuration
- One M.2 drive or one 2.5-inch drive (If OS is needed by debugging)
- Fifteen system fans
- One OCP module with required cable (if network is required)

Ranges for typical/peak loads are best guesstimate,
actual loads needs validation.

Three systems may draw approximately 27-36kw under typical load and up to 45-60kw under peak gpu utilization.

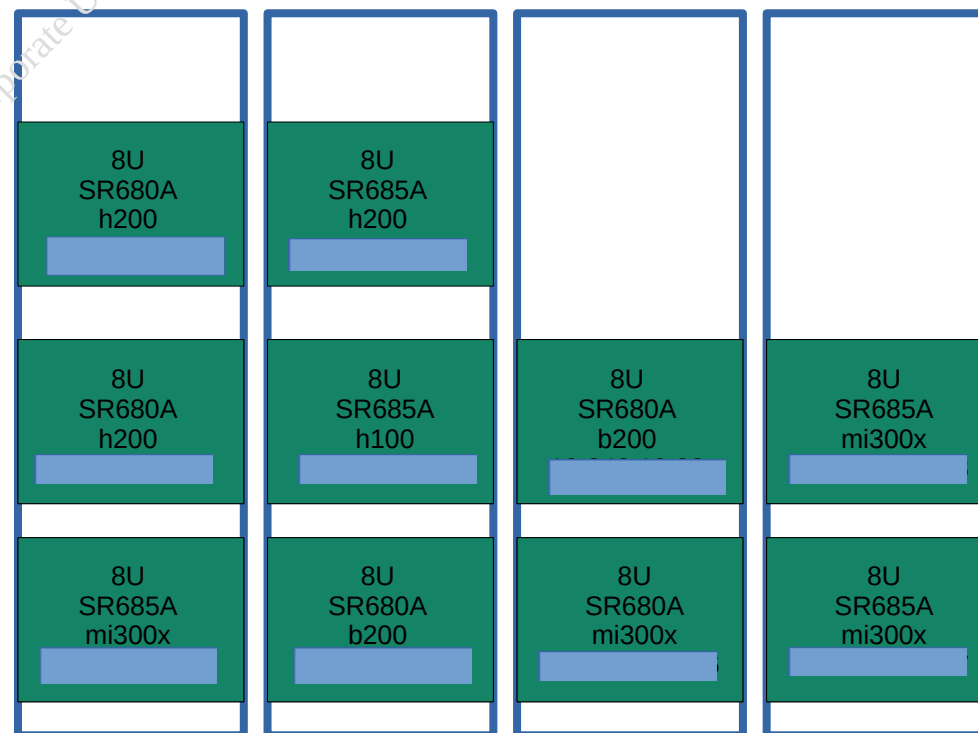
Base on estimated peak draw vs assumed drop capacity, the current 3 chassis, per rack configuration on the existing power infrastructure, any 'best effort' deployment in this environment appears significantly under-provisioned and carries the high risk of thermal shutdown, circuit failure, hardware damage, system instability.

In addition to the power infrastructure constraints, there's also physical constraints. The physical rack depth appears insufficient with these 8U chassis, potentially impacting airflow and pdu placement.

Model: Thinkserver SR680/685/780 8U chassis

Minimum power config: 6 psu x 2600w ~15.6kw max psu capacity

Recommended power config: 8 psu x 2600w ~20.8kw max psu capacity

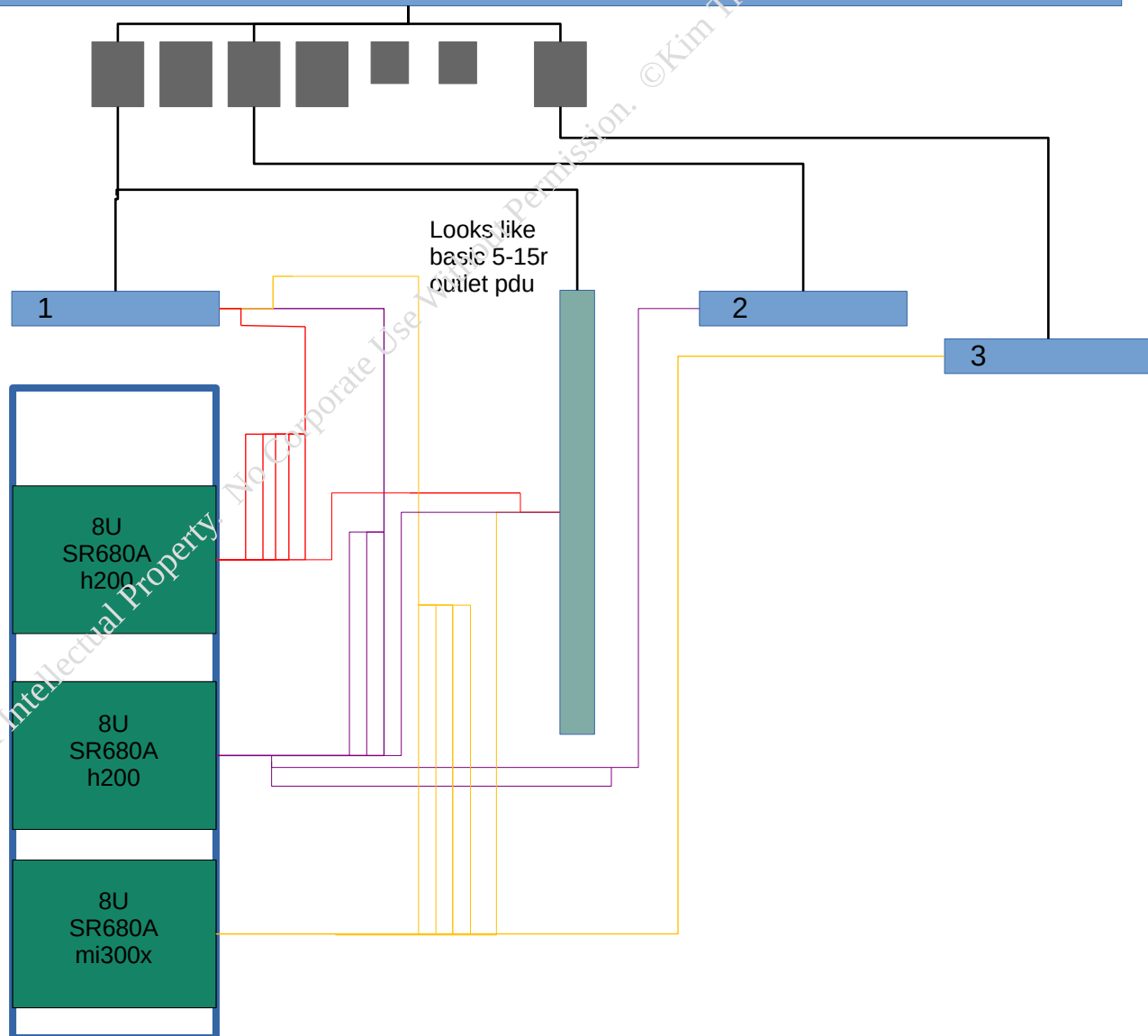


Actual physical layout as inventoried rack [1]

Any deviations means
change management done
without my awareness

225A 3PH 4W

1U 60A/208V 3P
C19 x 12 port PDU



Actual physical layout as inventoried rack [2]

Any deviations means
change management done
without my awareness

225A 3PH 4W

1U 60A/208V 3P
C19 x 12 port PDU

1

Looks like
basic 5-15r
outlet pdu

2

3

psu 1-6
1=1-1:2, 2=2:12
3=2:11, 4=2:10
5=2:9, 6=2:6

8U
SR685A
h200

psu 1-6
1=1-1:5, 2=2:8
3=3:4, 4=3:1
5=3:3, 6=3:2

8U
SR685A
h100

psu 1-5
1=2:3, 2=2:4
3=2:5, 4=2:1
5=2:2

8U
SR680A
b200

Actual physical layout as inventoried rack [3]

Any deviations means
change management done
without my awareness

225A 3PH 4W

Due to environment constraints,
connections were not traced
prior to re-work

1U 60A/208V 3P
C19 x 12 port PDU

1

Looks like
basic 5-15r
outlet pdu

2

3

8U
SR680A
b200

8U
SR680A
mi300x

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Actual physical layout as inventoried rack [4]

Any deviations means
change management done
without my awareness

225A 3PH 4W

Due to shared environment
with other equipment in rack,
connections not traced due to
environment constraints

1U 60A/208V 3P
C19 x 12 port PDU

1

Looks like
basic 5-15r
outlet pdu

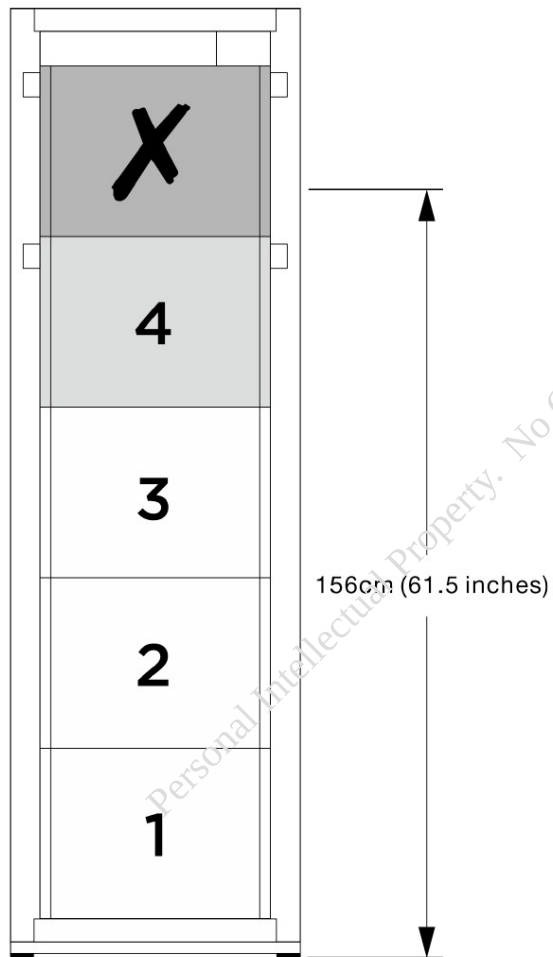
2

3

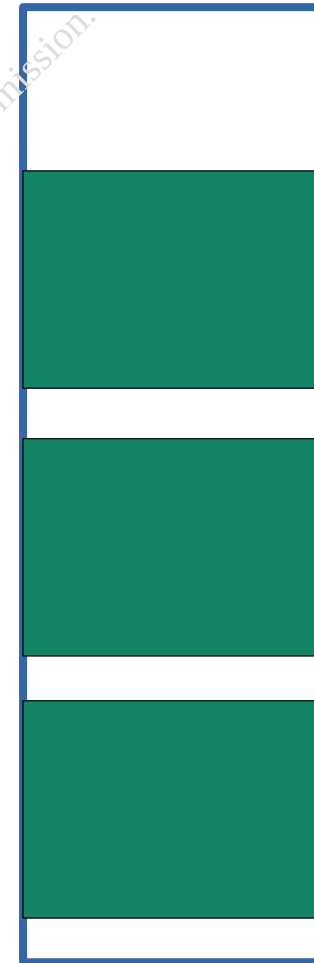
8U
SR680A
mi300x

8U
SR680A
mi300x

Per system documentation / best practices, 4 systems could be installed in rack, but the **assumption is that proper depth (deep) cabinet, and adequate power infrastructure to support the deployment**



Given the constraints of the lab infrastructure environment, 3 systems per rack would be ideal, but again the **assumption would be adequate power to support the deployment**



system recommended requirements
per system documentation/industry best practices

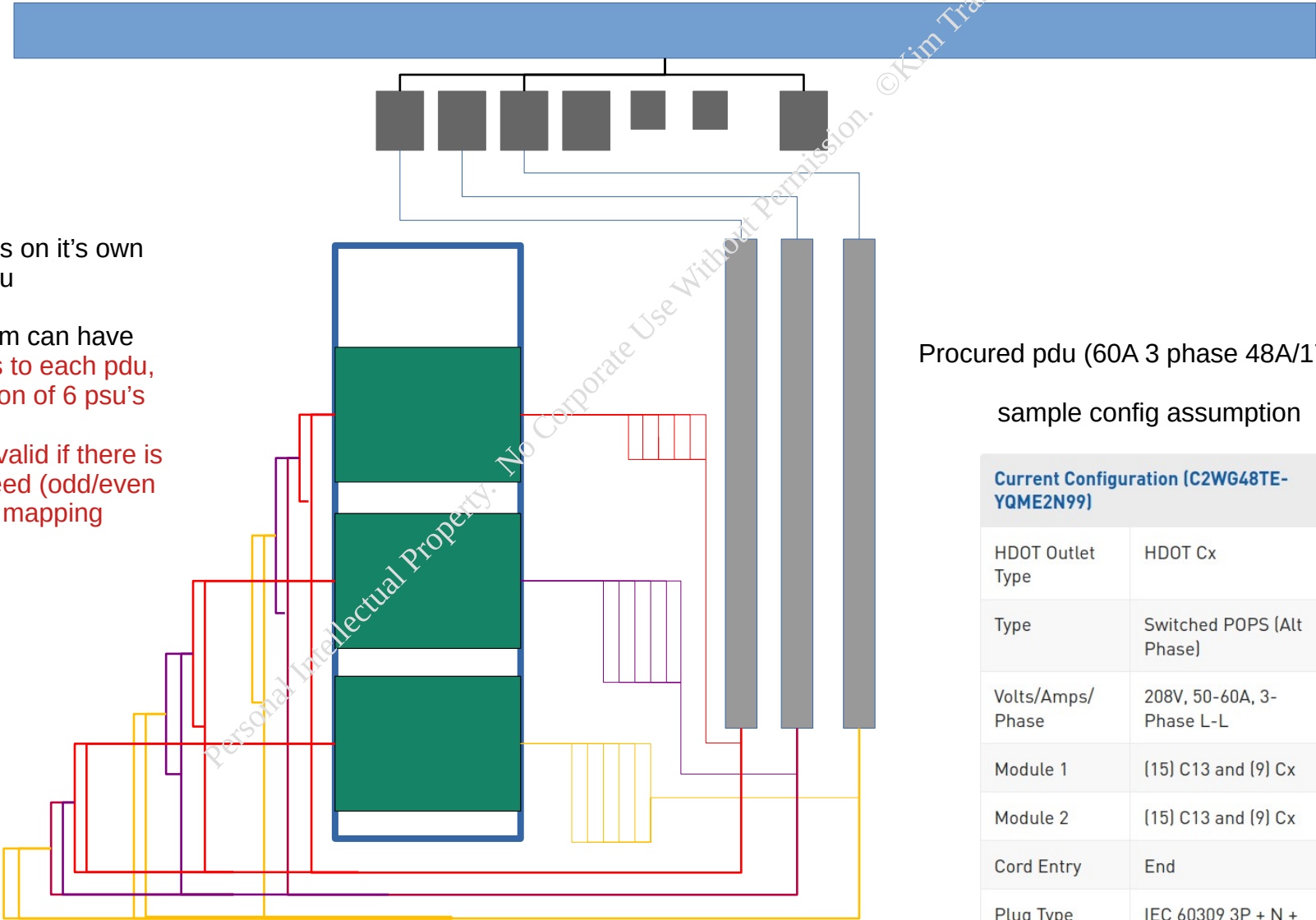
225A 3PH 4W

Ranges for typical/peak loads are best
guesstimate, actual loads needs validation.

each system is on it's own
pdu

or each system can have
2x connections to each pdu,
with assumption of 6 psu's

2x pdu is only valid if there is
alternate a/b feed (odd/even
psu to pdu mapping



Procured pdu (60A 3 phase 48A/17KW)

sample config assumption

Current Configuration (C2WG48TE-YQME2N99)	
HDOT Outlet Type	HDOT Cx
Type	Switched POPS (Alt Phase)
Volts/Amps/Phase	208V, 50-60A, 3-Phase L-L
Module 1	(15) C13 and (9) Cx
Module 2	(15) C13 and (9) Cx
Cord Entry	End
Plug Type	IEC 60309 3P + N + PE 60A

Given current lab infrastructure constraints: rack availability,
power drop availability, system workload...
Possible workable best effort deployment
RACK 1 RE-WORKED

Any deviations means
change management done
without my awareness

225A 3PH 4W

1U 60A/208V 3P
C19 x 12 port PDU

1

Looks like
basic 5-15r
outlet pdu

2

3

psu 1-6
1=1:1, 2=1:2
3=1:3, 4=1:4
5=1-1:6, 6=1-1:5

8U
SR680A
h200

psu 1-6
1=1:5, 2=1:6
3=1:7, 4=1:8
5=1-1:4, 6=1-1:3

8U
SR680A
h200

psu 1-6
1=1:9, 2=1:10
3=1:11, 4=1:12
5=1-1:2, 6=1-1:1

8U
SR680A
mi300x

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Given current lab infrastructure constraints: rack availability,
power drop availability, system workload...
Possible workable best effort deployment
RACK 2 RE-WORKED

Any deviations means
change management done
without my awareness

225A 3PH 4W

1U 60A/208V 3P
C19 x 12 port PDU

1

Looks like
basic 5-15r
outlet pdu

2

3

psu 1-6
1=2:1, 2=2:2
3=2:3, 4=2:4
5=2-1:6, 6=2-1:5

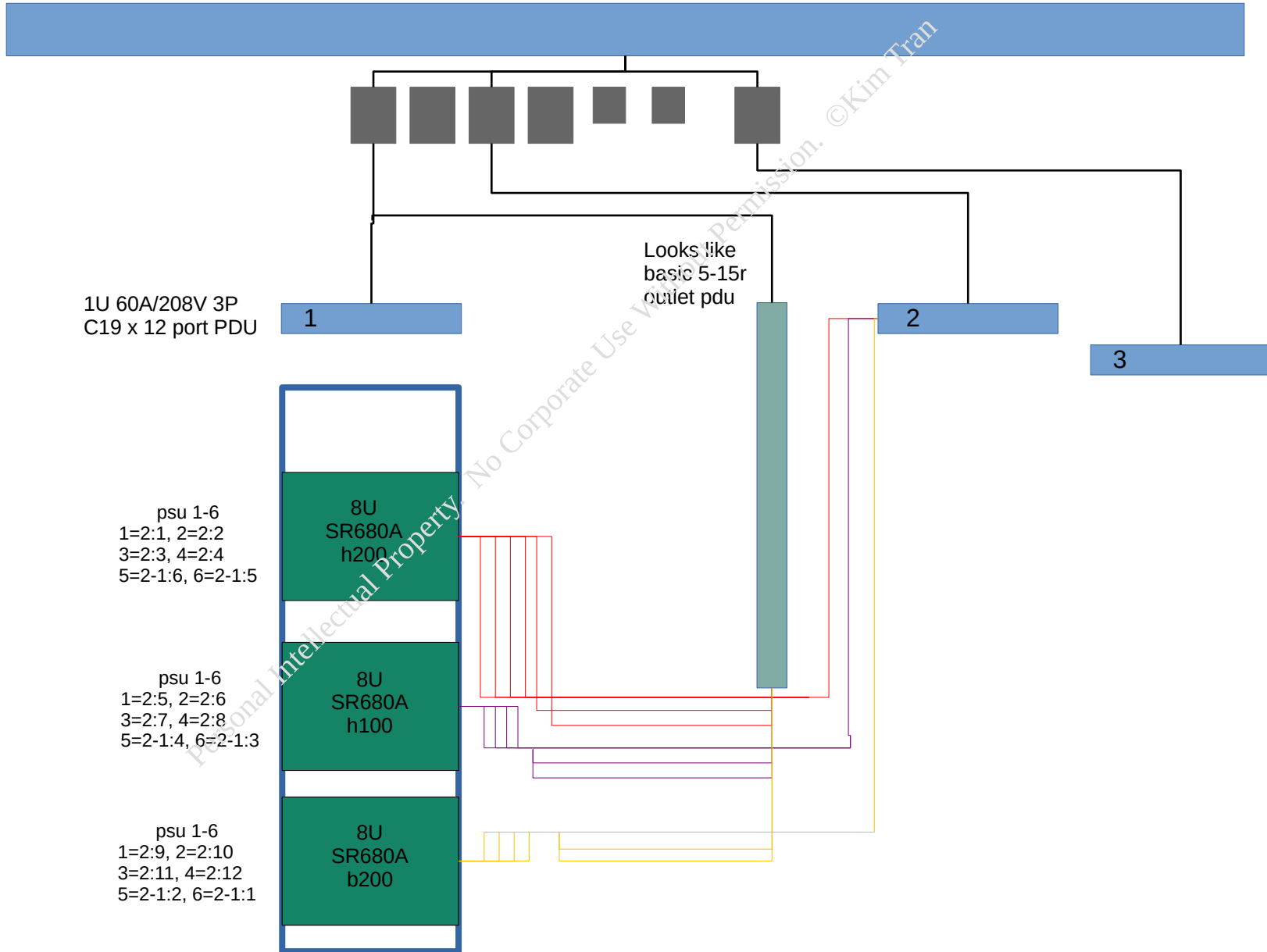
8U
SR680A
h200

psu 1-6
1=2:5, 2=2:6
3=2:7, 4=2:8
5=2-1:4, 6=2-1:3

8U
SR680A
h100

psu 1-6
1=2:9, 2=2:10
3=2:11, 4=2:12
5=2-1:2, 6=2-1:1

8U
SR680A
b200



Given current lab infrastructure constraints: rack availability,
power drop availability, system workload...
Possible workable best effort deployment
RACK 3 RE-WORKED

Any deviations means
change management done
without my awareness

225A 3PH 4W

1U 60A/208V 3P
C19 x 12 port PDU

1

Looks like
basic 5-15r
outlet pdu

2

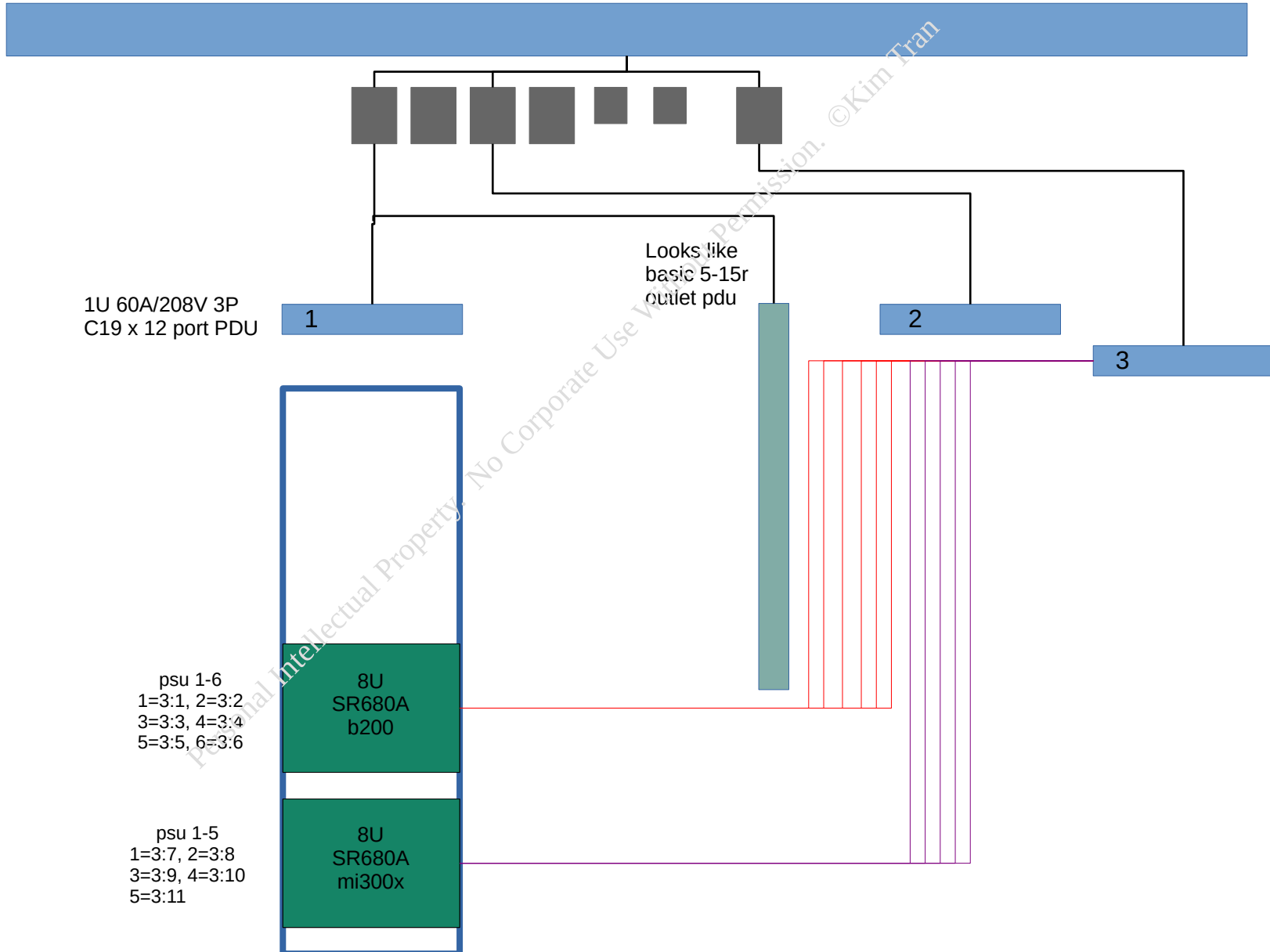
3

psu 1-6
1=3:1, 2=3:2
3=3:3, 4=3:4
5=3:5, 6=3:6

8U
SR680A
b200

psu 1-5
1=3:7, 2=3:8
3=3:9, 4=3:10
5=3:11

8U
SR680A
mi300x



Given current lab infrastructure constraints: rack availability,
power drop availability, system workload...
Possible workable best effort deployment
RACK 4 NOT RE-WORKED

Any deviations means
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without my awareness

225A 3PH 4W

Due to shared environment
with other equipment in rack,
connections not re-worked

1U 60A/208V 3P
C19 x 12 port PDU

1

Looks like
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outlet pdu

2

3

8U
SR680A
mi300x

8U
SR680A
mi300x

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Review the system reliability guidelines to ensure proper system cooling and reliability.

Make sure the following requirements are met:

When the server comes with redundant power, a power supply must be installed in each power-supply bay

Important: For any AI rack server that supports up to eight CFF PSUs in the rear of the server and operates with N+N redundancy, the two rack-level AC lines feeds must alternate between the PSUs to ensure balanced power distribution and redundancy.

- Rack-level AC line feed A: Connect to PSU 1, 3, 5, 7 (odd numbered PSUs)
- Rack-level AC line feed B: Connect to PSU 2, 4, 6, 8 (even numbered PSUs)

The following minimal configuration is required for the server to start:

- System board assembly that includes the processor board and system I/O board
- Firmware and RoT security module
- Two processors
- Twenty-four memory modules
- Six or eight power supplies, depending on the configuration
- One M.2 drive or one 2.5-inch drive (If OS is needed by debugging)
- Fifteen system fans
- One OCP module with required cable (if network is required)

Per system/platform documentation and best practices:

The server supports configurations with six or eight power supplies:

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Following is the list of supported type:

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Important: Power supplies and redundant power supplies in the server must be with the same power rating, wattage or level.

Minimal configuration for debugging

- System board assembly that includes the processor board and system I/O board
- Firmware and RoT security module
- Two processors
- Twenty-four memory modules
- Six or eight power supplies, depending on the configuration
- One M.2 drive or one 2.5-inch drive (If OS is needed by debugging)
- Fifteen system fans
- One OCP module with required cable (if network is required)

Adequate space around the server must be spared to allow server cooling system to work properly. Leave approximately 50 mm (2.0 in.) of open space around the front and rear of the server. Do not place any object in front of the fans.

Note: Even when power capping is disabled, the system may be throttled under certain fault conditions, such as power supply failure, cooling issue, etc.

Configuring the power capping policy

To configure the power capping policy, use the information in this topic.

Notes:

- AMD servers do not support configuring the power capping policy function.
- In an enclosure containing high-density server nodes, the chassis cooling and pow

When you install a new server, download and apply the latest firmware. This will help ensure that any known issues are addressed, and that your server is ready to work with optimal performance. Go to <https://datacentersupport.lenovo.com/tw/en/products/servers/thinksystem/sr680av3/7dhe/downloads/driver-list/> to download firmware updates for your server.